

CONTROLLED UNCLASSIFIED INFORMATION

Sole Source (Including Brand Name) Justification - Simplified Procedures for Certain Commercial Items

NOTE: If a Justification was approved for the preceding acquisition, a copy of the approved Justification for the predecessor action must be included in the staff package for approval of the instant Justification. This applies to Justification staff packages that are submitted for review and approval at a level above the contracting officer. The predecessor Justification will be used as a reference document by the approving official.

Is this a new or amended J&A Document? ☒ New ☐ Amended (Prior to Award Only!)

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☒ No

Funding level for this acquisition: [REDACTED]

Contracting Activity: AFTC/PZZF, Edwards AFB, CA 93524-1185

Purchase Request / Local ID Number: F150AG6015A001/ACQ-26-H002

Program / Project (and PE, if applicable): OPTICAMS Improvement & Modernization Project.

Program Type (PEO or Other Contracting): Other

Authority: ☒ AFFARS 5313.501 - 10 U.S.C. 2302b, Implementation of Simplified Acquisition Procedures (41 U.S.C. 1901)

☐ AFFARS 5313.501 - 10 U.S.C. 2304a, Special Emergency Procurement Authority (41 U.S.C. 1903)

Estimated Contract Cost [REDACTED]

Justification Type: ☐ Class ☒ Individual

COORDINATION ([AFFARS 5306.304\(a\)](#))

Date	Program Manager	[REDACTED]
12 Mar 2026	Justin Duhart	
	812 TSS/ENTI/ 661-277-4593	

[APPROVAL \(AFFARS 5306.304\(a\)\)](#)

Date	Contracting Officer	[REDACTED]
26 Mar 2026	Susana Dryer	
	AFTC/PZZF 661-277-2370	

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I. Contracting Activity.

The equipment that will be acquired is in support of the development of a EO/IR lab at Edwards AFB, Air Force Test Center (AFTC), Directorate of Contracting, Installation Contracting Division,

II. Nature and/or description of the action being processed.

This is a justification for a sole-source, purchase order award of a Firm-Fixed-Price contract to Santa Barbara Infrared, Inc. (SBIR) for the OPTICAMS Improvement & Modernization Project. This action is for the acquisition of a fully integrated, turnkey radiometric test station.

III. Description of supplies/services required to meet agency needs.

The required system is a single, cohesive, and fully integrated test station that functions as a turnkey system. The station is composed of three interdependent subsystems controlled by a single software interface for full automation:

High-Emissivity Blackbody Sources: Large-area (6"x6") flat-plate sources with a spectrally flat emissivity greater than 0.995 this is a major minimum requirement for the government.

Integrating Sphere Uniform Source: The specification require a reflective band uniform source, large Aperture Off-Axis Collimator and Control Software: The central controller (SBIR IRWindows™5) for the entire test setup.

Model 13971 Custom COLOSUS Target Projector System requires the following:

- T34B-.2833 4Bar Target for 3" target wheel with .2833" bar widths
- T34B-.2045 4Bar Target for 3" target wheel with .2045" bar widths
- T34B-.1278 4Bar Target for 3" target wheel with .1278" bar widths
- T34B-.1076 4Bar Target for 3" target wheel with .1076" bar widths
- T34B-.0043 4Bar Target for 3" target wheel with .0043 bar widths
- T34B-.0017 4Bar Target for 3" target wheel with .0017" bar widths
- T34B-.0011 4Bar Target for 3" target wheel with .0011" bar widths
- T34B-.0009 4Bar Target for 3" target wheel with .0009" bar widths
- Edge Target for 3" Target Wheel with 2.700" Edge Diameter
- T3P-.0426 Pinhole Target for 3" Target Wheel with .0426" Diameter Pinhole
- T3P-.3068 Pinhole Target for 3" Target Wheel with .3068" Diameter Pinhole

- 6" Differential Blackbody (0°C to 175°C Abs T, -25°C to +150°C Delta T) with: Controller, Blackbody Head, 10' Cable, Ethernet, GPIB & RS-232 with VANTABLACK S-IR.

- 12" Differential Blackbody (0C to 175C Abs T, -25C to +150C Delta T) with: Controller, Blackbody Head, 10' Cable, Ethernet, GPIB & RS-232 with VANTABLACK S-IR

- 8" Extended Area Low Temp Blackbody (-40°C to 100°C) with: Controller, Blackbody Head, Chiller, 10' Cable, Ethernet, GPIB & RS-232 with VANTABLACK S-IR

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-GFE: Optical table for system installation

- 6" High Temperature Blackbody System (+50°C to +600°C Absolute T) with: Controller, Blackbody Head, 10' Cable and GPIB Interface

- Annual Support Subscription (IRWindows 5) includes: priority response within 24 hours via dedicated email address for up to 12 hours of technical support & up to 2 software release updates per year containing new features & upgrades.

IV. Statutory authority permitting other than full and open competition.

The statutory authority permitting other than full and open competition for this acquisition is 10 U.S.C. § 3204(a)(1), as implemented by the Federal Acquisition Regulation (RFO) Part 6.103-1, "Only one responsible source and no other supplies or services will satisfy agency requirements."

This justification has been prepared in full compliance with the procedural requirements of RFO Subpart 6.103-1, as supplemented by all applicable agency regulations, including the requirements outlined in Class Deviation 2026-O0017 regarding competition requirements.

V. Demonstration that the contractor's unique qualifications or nature of the acquisition requires the use of the authority cited above (applicability of authority).

The Air Force Test Center (AFTC) has a mission-critical requirement for a new, turnkey Electro-Optical/Infrared (EO/IR) Target Projector System. This system is essential for the laboratory characterization, performance verification, and calibration of advanced sensor systems. The Government requires a single, fully integrated system from one vendor to eliminate technical risk associated with integrating disparate components and to ensure data integrity through a unified software control platform.

Specific Requirements:

1. SR-1 System Performance; The system shall be capable of projecting targets across the visible spectrum through the Long-Wave Infrared (LWIR) spectrum (approximately 0.4µm to 14µm).
2. SR-2 Radiometric Accuracy; The system must include at least two blackbody infrared sources, each demonstrating a minimum emissivity of > 0.995. One source shall be a differential type, and the other shall be an absolute type.
3. SR-3 Visible/SWIR Capability; The system must include an integrated uniform source (e.g., an integrating sphere) for sensor testing in the visible through Short-Wave Infrared (SWIR) spectrum.
4. SR-4 Hardware Integration; The system must be delivered as a complete, factory-integrated turnkey solution, including all sources, collimator, motorized target wheels, and system controller rack. The Government will not act as a systems integrator.
5. SR- 5 Software Integration; All hardware components must be controllable from a single, unified software interface provided with the system. The software shall be delivered fully installed, configured, and capable of executing automated test sequences.

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6. SR-6 Support Services; The vendor shall provide on-site installation, final system alignment, and complete functional verification at the Government facility.

7. SR-7 Training; The vendor shall provide on-site operator training for up to eight (8) Government personnel, covering system operation, maintenance, and use of the control software.

8. SR-8 Warranty: Provide standard commercial warranty to commence upon Government acceptance.

9. Blackbodies and collimator, electro-optical test equipment with blackbody technology, utilizing ultra-low reflectance coatings to achieve emissivity values approaching 100%.

10. VANTABLACK®S-IR versatile coating based upon similar CNT technology. The application process results in a coating with enhanced absorption properties and a passive surface that repels water and other contaminants.

11. Visible spectrum (less than 0.2% at 700nm), SWIR, MWIR and LWIR spectral ranges. The coating has been shown to have good resistance to water adsorption, thermal shock, mechanical shock and vibration.

SBIR is the only source capable of meeting the Government's requirements for both technical performance and critical system interoperability.

The Original Equipment Manufacturer (OEM) of Proprietary Legacy System: This acquisition is a modernization of a long-standing Government investment in a legacy test system manufactured by SBIR under FA930224P0050, FA930225P0016. This system is fundamentally built upon SBIR's proprietary "IRWindows 5" software ecosystem, for which the Government does not possess the source code or data rights. The new components must be 100% compatible and fully interoperable with this existing proprietary software.

Unique Technical Capability: The Government has a mandatory requirement for large-area blackbody sources with a spectrally flat emissivity greater than 0.995. SBIR is the only identified source capable of meeting this need via its exclusive commercial license to apply the Vantablack S-IR coating to this type of equipment.

Using another source would result in the acquisition of an incompatible, technically unacceptable system, necessitating a high-risk, and substantial duplication of cost and reverse engineering effort that would fail to meet the Government's needs. As the government has invested on earlier irradiation of "IRWindows 5" FA865124CB030.

VI. Description of efforts made to ensure that offers are solicited from as many potential sources as practicable.

A comprehensive market survey was conducted in accordance with RFO Part 10. The primary effort was the issuance of a public Sources Sought Notice (Notice ID: F1S0AG6015A001) on SAM.gov. All responses to this notice were formally reviewed by the Government's technical team to assess their capabilities against the Government's minimum needs. This public notice and the subsequent

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analysis confirmed that no other sources could meet the mandatory requirements.

VII. Determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

The determination will be based on a price analysis in accordance with RFO 15.404-1(b). The primary technique will be a comparison of the contractor's proposal with their established commercial pricing.

VIII. Description of the market research conducted and the results, or a statement of the reasons market research was not conducted.

Market research confirmed that a sole-source award is the only viable strategy. The research included a public Sources Sought Notice on SAM.gov, a formal technical analysis of all responses received (from SBIR, Federon/HGH, and AFRL), and a review of historical acquisitions (e.g., contract FA865124CB030). The detailed findings are documented in the associated Market Research Report (MRR), which concludes that no other source can meet the Government's minimum needs.

IX. Any other facts supporting the use of Other Than Full and Open Competition.

In accordance with RFO 6.103-1(a)(2)(i) Only one responsible source and no other supplies or services will satisfy agency requirements, an award to any other source would result in substantial and quantifiable duplication of cost to the Government. The Government's investment in the existing AFTC test capability under contract FA865124CB030 represents a significant sunk cost. Based on the initial value of that acquisition and subsequent investments in software maintenance, it is estimated that introducing a new, non-compatible system would require the Government to abandon a capability with an approximate replacement value of [REDACTED]

This figure represents the estimated cost to acquire a completely new, non-SBIR system with similar capabilities [REDACTED] plus the associated costs for new operator and maintenance training, facility modifications, and the development of new test procedures [REDACTED]. These duplicated costs, which do not include the unacceptable schedule delays and mission risk, would not be recovered through competition for this specific follow-on requirement.

X. List of any sources that expressed, in writing, an interest in the acquisition.

Federon, LLC (on behalf of HGH Infrared Systems) and Air Force Research Laboratory (AFRL/ITWL).

XI. A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before making subsequent acquisitions for the supplies or services required.

The primary barrier to competition for this requirement is the proprietary nature of the Government's existing IRWindows software, which is integral to the legacy system acquired under contract FA865124CB030. The Government does not own the data rights to this software, precluding other vendors from developing compatible hardware. Due to this embedded

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proprietary technology, no immediate actions can be taken to increase competition for this specific modernization requirement. However, for future, next-generation system acquisitions, the agency will conduct market research to determine if modular, open-system architectures have become available, which would allow for component-based competition and remove the current proprietary barrier.

XII. Certification by the Contracting Officer.

As evidenced by my signature above, I have determined this document to be both accurate and complete to the best of my knowledge and belief.

XIII. Certification by the technical/requirements personnel.

As evidenced by my (our) signature(s) above, I (we) certify that any supporting data contained herein, which is my (our) responsibility, is both accurate and complete